

Welding Procedure Specification (WPS) — Aluminum MIG Welding

- 1. Process:** - GMAW (Gas Metal Arc Welding — MIG)
- 2. Base Material:** - Material: Aluminum - Alloy: 6061-T6 (or similar) - Thickness Range: 1/8" to 1/2"
- 3. Joint Design:** - Butt, lap, and T-joints - Groove preparation as needed - Fit-up: tight with minimal gaps
- 4. Filler Metal:** - ER4043 or ER5356 - Diameter: 0.030" or 0.035"
- 5. Shielding Gas:** - 100% Argon - Flow Rate: 20–30 CFH
- 6. Preheat and Interpass Temp:** - Preheat: Not required for thin material (<1/4"); 200–300°F for thicker sections - Interpass Temp: <300°F
- 7. Electrical Parameters:** - Polarity: DCEP (Direct Current Electrode Positive) - Transfer Mode: Spray or Pulsed Spray (preferred for clean, strong welds) - Voltage: 18–28 V (varies with material thickness) - Wire Feed Speed: 250–500 IPM (adjust per wire diameter and material)
- 8. Technique:** - Stringer or slight weave (avoid wide weaving) - Push technique (gun angle ~10–15°) - Travel Speed: Medium to fast to prevent burn-through - Keep tip-to-work distance ~3/4"
- 9. Equipment:** - Spool gun or push-pull feeder recommended - Clean liner, contact tip, and drive rolls for aluminum
- 10. Cleaning & Prep:** - Mechanically or chemically remove oxide layer before welding - Degrease with acetone or approved solvent
- 11. Post-Weld Operations:** - Visual inspection for cracks, porosity, and undercut - Optional: X-ray or dye penetrant for critical welds - Remove spatter and clean weld area
- 12. Safety:** - Proper PPE: welding helmet, gloves, apron - Ventilation for aluminum fume control

Note: Always verify procedure with job-specific codes or AWS D1.2 standards if required.